Problem 1:

A) Navigate and list :

a. Navigate and List: a. Start by navigating to your home directory and list its contents. Then, move into a directory named "LinuxAssignment" if it exists; otherwise, create it.

suchit@DESKTOP-DJ82803:~$ mkdir LinuxAssignment

suchit@DESKTOP-DJ82803:~$ cd LinuxAssignment

suchit@DESKTOP-DJ82803:~/LinuxAssignment$

```
suchit@DESKTOP-DJ82803:~$ ls
ad.txt  com.sh  grandfather  ls  user1  user2
suchit@DESKTOP-DJ82803:~$ mkdir LinuxAssignment
suchit@DESKTOP-DJ82803:~$ cd LinuxAssignment/
suchit@DESKTOP-DJ82803:~/LinuxAssignment$
```

B) File Management :

a. File Management: a. Inside the "LinuxAssignment" directory, create a new file named "file1.txt". Display its contents.

suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cat >file1.txt

Hello

suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cat file1.txt

Hello

```
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cat >file1.txt
Hello
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cat file1.txt
Hello
suchit@DESKTOP-DJ82803:~/LinuxAssignment$
```

C) Directory Management :

a. Create a new directory named "docs" inside the "LinuxAssignment" directory.

suchit@DESKTOP-DJ82803:~/LinuxAssignment$ mkdir docs

suchit@DESKTOP-DJ82803:~/LinuxAssignment$ ls

docs file1.txt

```
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ mkdir docs
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ ls
docs  file1.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment$
```

D) Copy and move files :

a. Copy the "file1.txt" file into the "docs" directory and rename it to "file2.txt".

suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cp file1.txt docs/file2.txt

suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cd docs/

suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$ ls

file2.txt

suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$

```
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cp file1.txt docs/file2.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cd docs/
suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$ ls
file2.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$
```

E) Permissions and Ownerships :

a. Change the permissions of "file2.txt" to allow read, write, and execute permissions for the owner and only read permissions for others. Then, change the owner of "file2.txt" to the current user

```
suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$ ls -l
total 4
-rw-r--r-- 1 suchit suchit 6 Mar  7 18:11 file2.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$ chmod u+x file2.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$ ls -l
total 4
-rwxr--r-- 1 suchit suchit 6 Mar  7 18:11 file2.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$ chown suchit file2.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$ ls -a -l
total 12
drwxr-xr-x 2 suchit suchit 4096 Mar  7 18:11 .
drwxr-xr-x 3 suchit suchit 4096 Mar  7 18:09 ..
-rwxr--r-- 1 suchit suchit    6 Mar  7 18:11 file2.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$ sudo su
[sudo] password for suchit:
root@DESKTOP-DJ82803:/home/suchit/LinuxAssignment/docs# ls -a -l
total 12
drwxr-xr-x 2 suchit suchit 4096 Mar  7 18:11 .
drwxr-xr-x 3 suchit suchit 4096 Mar  7 18:09 ..
-rwxr--r-- 1 suchit suchit    6 Mar  7 18:11 file2.txt
root@DESKTOP-DJ82803:/home/suchit/LinuxAssignment/docs# su os
su: user os does not exist or the user entry does not contain all the re
quired fields
root@DESKTOP-DJ82803:/home/suchit/LinuxAssignment/docs#
```

F) Final Checklist :

a. Finally, list the contents of the "LinuxAssignment" directory and the root directory to ensure that all operations were performed correctly.

```
suchit@DESKTOP-DJ82803:~/LinuxAssignment/docs$ cd
suchit@DESKTOP-DJ82803:~$ ls LinuxAssignment/
docs  file1.txt
suchit@DESKTOP-DJ82803:~$ tree LinuxAssignment/
LinuxAssignment/
├── docs
│   └── file2.txt
└── file1.txt

1 directory, 2 files
suchit@DESKTOP-DJ82803:~$
```

G) File Searching :

a. Search for all files with the extension ".txt" in the current directory and its subdirectories.

```
suchit@DESKTOP-DJ82803:~$ find -name "*.txt"
./LinuxAssignment/file1.txt
./LinuxAssignment/docs/file2.txt
./ad.txt
suchit@DESKTOP-DJ82803:~$
```

b. Display lines containing a specific word in a file (provide a file name and the specific word to search).

```
suchit@DESKTOP-DJ82803:~$ cat >abc.txt
This is a new file
we want to find a word in here
my name is suchit
suchit is my name
suchit@DESKTOP-DJ82803:~$ grep "suchit" abc.txt
my name is suchit
suchit is my name
suchit@DESKTOP-DJ82803:~$
```

H) System Information :

a. Display the current system date and time.

```
suchit@DESKTOP-DJ82803:~$ date
Thu Mar  7 19:01:48 IST 2024
suchit@DESKTOP-DJ82803:~$
```

I) Networking :

a. Display the IP address of the system.

```
suchit@DESKTOP-DJ82803:~$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN grou
p default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc mq state UP gr
oup default qlen 1000
    link/ether 00:15:5d:cf:17:8e brd ff:ff:ff:ff:ff:ff
    inet 192.168.220.233/20 brd 192.168.223.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fecf:178e/64 scope link
       valid_lft forever preferred_lft forever
suchit@DESKTOP-DJ82803:~$
```

b. Ping a remote server to check connectivity (provide a remote server address to ping).

```
suchit@DESKTOP-DJ82803:~$ ping www.google.com
PING www.google.com (216.239.38.10) 56(84) bytes of data.
64 bytes from ns4.google.com (216.239.38.10): icmp_seq=1 ttl=58 time=67.4 ms
64 bytes from ns4.google.com (216.239.38.10): icmp_seq=2 ttl=58 time=82.6 ms
64 bytes from ns4.google.com (216.239.38.10): icmp_seq=3 ttl=58 time=109 ms
64 bytes from ns4.google.com (216.239.38.10): icmp_seq=4 ttl=58 time=134 ms
64 bytes from ns4.google.com (216.239.38.10): icmp_seq=5 ttl=58 time=65.9 ms
64 bytes from ns4.google.com (216.239.38.10): icmp_seq=6 ttl=58 time=187 ms
64 bytes from ns4.google.com (216.239.38.10): icmp_seq=7 ttl=58 time=110 ms
^C
--- www.google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6012ms
rtt min/avg/max/mdev = 65.938/107.999/186.996/39.644 ms
suchit@DESKTOP-DJ82803:~$
```

J) File Compression :

a. Compress the "docs" directory into a zip file.

```
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ ls
docs  file1.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ zip docs.zip docs
  adding: docs/ (stored 0%)
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ ls
docs  docs.zip  file1.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment$
```

b. Extract the contents of the zip file into a new directory.

```
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ ls
docs  file1.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ zip docs.zip docs
  adding: docs/ (stored 0%)
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ ls
docs  docs.zip  file1.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ mkdir new
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ mv docs.zip new
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cd new/
suchit@DESKTOP-DJ82803:~/LinuxAssignment/new$ unzip docs.zip
Archive:  docs.zip
   creating: docs/
suchit@DESKTOP-DJ82803:~/LinuxAssignment/new$ ls
docs  docs.zip
suchit@DESKTOP-DJ82803:~/LinuxAssignment/new$
```

K) File Editing :

a. Open the "file1.txt" file in a text editor and add some text to it.

b. Replace a specific word in the "file1.txt" file with another word (provide the original word and the word to replace it with).

```
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ ls
docs  file1.txt  new
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cat file1.txt
Hello
The quick brown fox jumps over the lazy dog.
Lorem ipsum dolor sit amet, consectetur adipiscing elit.
This is some sample text for demonstration purposes.
Hello, world! How are you today?
Programming is fun and challenging.

suchit@DESKTOP-DJ82803:~/LinuxAssignment$ tr 'Hello' 'Hi' < file1.txt >
temp_file && mv temp_file file1.txt
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ ls
docs  file1.txt  new
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cat fie
cat: fie: No such file or directory
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ file
file      filefrag
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ file
file      filefrag
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ ls
docs  file1.txt  new
suchit@DESKTOP-DJ82803:~/LinuxAssignment$ cat file1.txt
Hiiii
Thi quick briwn fix jumps ivir thi iazy dig.
Lirim ipsum diiir sit amit, cinsictitur adipiscing iiit.
This is simi sampii tixt fir diminstratiin purpisis.
Hiiii, wirid! Hiw ari yiu tiday?
Prigramming is fun and chaiiinging.

suchit@DESKTOP-DJ82803:~/LinuxAssignment$
```

Problem 2 :

a. Suppose you have a file named "data.txt" containing important information. Display the first 10 lines of this file to quickly glance at its contents using a command.

```
suchit@DESKTOP-DJ82803:~$ vi data.txt
suchit@DESKTOP-DJ82803:~$ ls
LinuxAssignment  ad.txt  data.txt     ls     user2
abc.txt          com.sh  grandfather  user1
suchit@DESKTOP-DJ82803:~$ head -n 20 data.txt
1. The quick brown fox jumps over the lazy dog.
2. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
3. Hello, world! How are you today?
4. This is some sample text for demonstration purposes.
5. Programming is fun and challenging.
6. Life is like riding a bicycle. To keep your balance, you must keep mo
ving.
7. In the end, we only regret the chances we didn't take.
8. Happiness is not something ready-made. It comes from your own actions
.
9. The only way to do great work is to love what you do.
10. Believe you can and you're halfway there.
11. The greatest glory in living lies not in never falling, but in risin
g every time we fall.
12. Success is not final, failure is not fatal: It is the courage to con
tinue that counts.
13. Everything you can imagine is real.
14. The only limit to our realization of tomorrow will be our doubts of
today.
15. It's not whether you get knocked down, it's whether you get up.
16. Strive not to be a success, but rather to be of value.
17. Life is 10% what happens to us and 90% how we react to it.
18. What you get by achieving your goals is not as important as what you
 become by achieving your goals.
19. When one door of happiness closes, another opens; but often we look
so long at the closed door that we do not see the one which has been ope
ned for us.
20. The only source of knowledge is experience.
suchit@DESKTOP-DJ82803:~$
```

b. Now, to check the end of the file for any recent additions, display the last 5 lines of "data.txt" using another command.

```
suchit@DESKTOP-DJ82803:~$ tail -n 5 data.txt
47. Believe you can and you're halfway there.
48. Success is not final, failure is not fatal: It is the courage to con
tinue that counts.
49. Your time is limited, don't waste it living someone else's life.
50. Life is what happens when you're busy making other plans.

suchit@DESKTOP-DJ82803:~$
```

c. In a file named "numbers.txt," there are a series of numbers. Display the first 15 lines of this file to analyze the initial data set.

```
suchit@DESKTOP-DJ82803:~$ vi numbers.txt
suchit@DESKTOP-DJ82803:~$ head -n 15 numbers.txt
10
20
30
40
5
2
3
5
2
0
52
8
4
2
6
suchit@DESKTOP-DJ82803:~$
```

d. To focus on the last few numbers of the dataset, display the last 3 lines of "numbers.txt".

```
suchit@DESKTOP-DJ82803:~$ vi numbers.txt
suchit@DESKTOP-DJ82803:~$ head -n 15 numbers.txt
10
20
30
40
5
2
3
5
2
0
52
8
4
2
6
suchit@DESKTOP-DJ82803:~$ tail -n 3 numbers.txt
6586


suchit@DESKTOP-DJ82803:~$ tail -n 5 numbers.txt
68
65
6586


suchit@DESKTOP-DJ82803:~$ vi numbers.txt
suchit@DESKTOP-DJ82803:~$ tail -n 3 numbers.txt
68
65
6586
suchit@DESKTOP-DJ82803:~$
```

e. Imagine you have a file named "input.txt" with text content. Use a command to translate all lowercase letters to uppercase in "input.txt" and save the modified text in a new file named "output.txt."

```
suchit@DESKTOP-DJ82803:~$ vi input.txt
suchit@DESKTOP-DJ82803:~$ cat input.txt

Hii everyone,
my name is suchit chaudhari
i am an electronics engineer.
i completed my B.Tech from Pillai College of Engineering
suchit@DESKTOP-DJ82803:~$ tr [:upper:] input.txt > output.txt
suchit@DESKTOP-DJ82803:~$ ls
LinuxAssignment  ad.txt  data.txt     input.txt  numbers.txt  user1
abc.txt          com.sh  grandfather  ls         output.txt   user2
suchit@DESKTOP-DJ82803:~$ cat output.txt
suchit@DESKTOP-DJ82803:~$ vi output.txt
suchit@DESKTOP-DJ82803:~$ tr [:lower:] [:upper:] <input.txt> temp_file && mv temp_file output.txt
suchit@DESKTOP-DJ82803:~$ cat output.txt

HII EVERYONE,
MY NAME IS SUCHIT CHAUDHARI
I AM AN ELECTRONICS ENGINEER.
I COMPLETED MY B.TECH FROM PILLAI COLLEGE OF ENGINEERING
suchit@DESKTOP-DJ82803:~$
```

f. In a file named "duplicate.txt," there are several lines of text, some of which are duplicates. Use a command to display only the unique lines from "duplicate.txt."

```
suchit@DESKTOP-DJ82803:~$ cat duplicate.txt
The world is full of wonders waiting to be discovered.
History teaches us valuable lessons about the past.
History teaches us valuable lessons about the past.
Science and technology continue to advance at a rapid pace.
Art and creativity enrich our lives in countless ways.
Art and creativity enrich our lives in countless ways.
Nature provides endless beauty and inspiration.
Exploring new cultures broadens our perspective on the world.
Education is the key to unlocking opportunities and growth.
Kindness and empathy make the world a better place for everyone.
Music has the power to evoke emotions and unite people.
Dreams have the potential to become reality with dedication and persever
ance.

suchit@DESKTOP-DJ82803:~$ uniq duplicate.txt
The world is full of wonders waiting to be discovered.
History teaches us valuable lessons about the past.
Science and technology continue to advance at a rapid pace.
Art and creativity enrich our lives in countless ways.
Nature provides endless beauty and inspiration.
Exploring new cultures broadens our perspective on the world.
Education is the key to unlocking opportunities and growth.
Kindness and empathy make the world a better place for everyone.
Music has the power to evoke emotions and unite people.
Dreams have the potential to become reality with dedication and persever
ance.

suchit@DESKTOP-DJ82803:~$
```

g. In a file named "fruit.txt," there is a list of fruits, but some fruits are repeated. Use a command to display each unique fruit along with the count of its occurrences in "fruit.txt."

```
suchit@DESKTOP-DJ82803:~$ cat >fruit.txt
apple
apple
orange
apple
grapes
orange
kiwi
kiwi
mango
papayasuchit@DESKTOP-DJ82803:~$
suchit@DESKTOP-DJ82803:~$ uniq -c fruit.txt
      2 apple
      1 orange
      1 apple
      1 grapes
      1 orange
      2 kiwi
      1 mango
      1 papaya
suchit@DESKTOP-DJ82803:~$
```